**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 159 · Paper 2: Searching and Sorting

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Why should a performance explanation go beyond the word faster?

Q2. A = 1 and B = 0. What is A AND B?
A. 1
B. 2
C. 0
D. 10

Q3. A circuit has X = A AND B, followed by Q = X OR C. For input rows 000, 011, 110, 111, give the pairs (X,Q). Why is a column for X useful?

Q4. RAM is full. Apps keep moving data between RAM and an SSD. Correct both claims. Explain why. (a) Virtual memory needs no secondary storage. (b) Virtual memory is faster than RAM because SSDs hold more.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Linear search — Checks elements sequentially, regardless of order.

Binary search — Requires sorted data and repeatedly halves active search range.

Bubble sort — Adjacent comparisons/swaps move large elements right over passes.

Insertion and merge — Insertion grows a sorted prefix; merge sorts sublists before combining.

## New learning

Linear search: Checks elements sequentially, regardless of order.

Binary search: Requires sorted data and repeatedly halves active search range.

Bubble sort: Adjacent comparisons/swaps move large elements right over passes.

Insertion and merge: Insertion grows a sorted prefix; merge sorts sublists before combining.

## Worked example

Trace bubble-sort first pass of [6,1,4,2].

1. Compare 6/1 swap → [1,6,4,2].

2. Compare 6/4 swap → [1,4,6,2].

3. Compare 6/2 swap → [1,4,2,6].

4. Conclude 6 has reached the end.

# We Do · Practise with guidance

### Guided 1

Use binary search to find 17 in [3,7,11,17,23].

Worked method: Found index3.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Finish the comparison that detects an out-of-order pair of file sizes.

```
leftGB = 6
rightGB = 1
if ... then
    print("Swap")
endif
```

write

### Question 1

Give the array after one bubble pass on [4,3,2,1]. [3 marks]

### Question 2

State the prerequisite for binary search. [1 mark]

### Question 3

Explain insertion sort in two stages. [3 marks]

# Exit Ticket · Answer before checking

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Binary search reliably finds targets in any unsorted list.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. It should identify the relevant characteristic and explain its effect on the stated task.

Q2. C — 0

Q3. (0,0), (0,1), (1,1), (1,1). X shows the first gate's output, helping work out Q and find errors.

Q4. (a) Virtual memory uses secondary storage when RAM is too full. (b) Moving data between RAM and slower secondary storage takes time. More storage space does not mean faster access.

### I Do / We Do pages · Guided practice

Q1. Found index3.

### You Do pages · Code First

```
leftGB = 6
rightGB = 1
if leftGB > rightGB then
    print("Swap")
endif
```

Expected result: Swap

### You Do pages · Independent questions

Q1. [3,2,1,4].

Q2. The list must be sorted.

Q3. Take next key from unsorted section; shift greater prefix elements right and insert key correctly.

### Exit Ticket · Model answer

False

# Try the question again

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**